Karnataka Law Society's

GOGTE INSTITUTE OF TECHNOLOGY

Belagavi

Project Expo-2024

Department of Information Science and Engineering

Project Title: Accident Detection And Alert System Using IoT

Motivation and Objectives:

Objectives:

- Design a system that can detect real accidents and ignore non-emergency events .
- Avoid false alarms and ensure that no real accidents go unreported.
- By leveraging IoT technology, this system revolutionizes accident response by providing instantaneous alerts to emergency services, potentially saving countless lives through swift intervention.
- Its seamless integration of sensors and communication modules marks a crucial step towards a safer and more efficient road safety infrastructure.

Motivation:

- By leveraging IoT technology, this system revolutionizes accident response by providing instantaneous alerts to emergency services, potentially saving countless lives through swift intervention. Its seamless integration of sensors and communication modules marks a crucial step towards a safer and more efficient road safety infrastructure.

Methodology :

- When it comes to road safety, effective and rapid accident response is crucial to reduce fatal injuries and save lives. Unfortunately, delays in reporting accidents to authorities can be fatal.
- The challenge is to design a system that accurately detects real accidents while avoiding false alarms, ensuring no incident goes unreported.
- Recent advancements leverage accelerometers and gyroscope sensors with machine learning to distinguish between normal conditions and accidents effectively.
- Integration of GPS and GSM technologies enables immediate location transmission to emergency responders, reducing response times. Additionally, IoT connectivity ensures continuous monitoring and immediate response upon accident detection.
- To minimize false alarms, sophisticated algorithms analyze historical data to learn from false triggers, enhancing system accuracy and reliability.

Project Group | Batch no. 06

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2023-24

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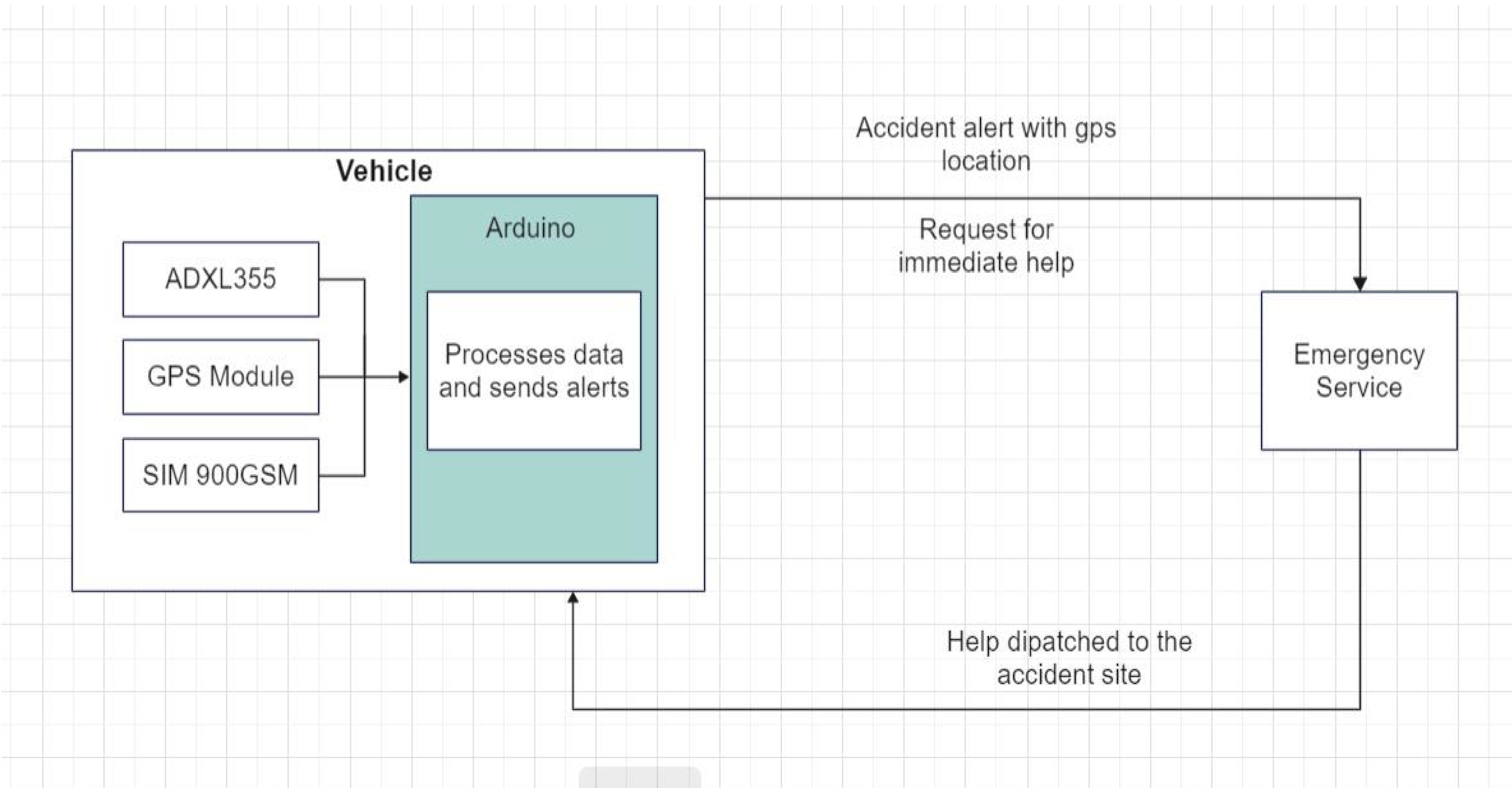
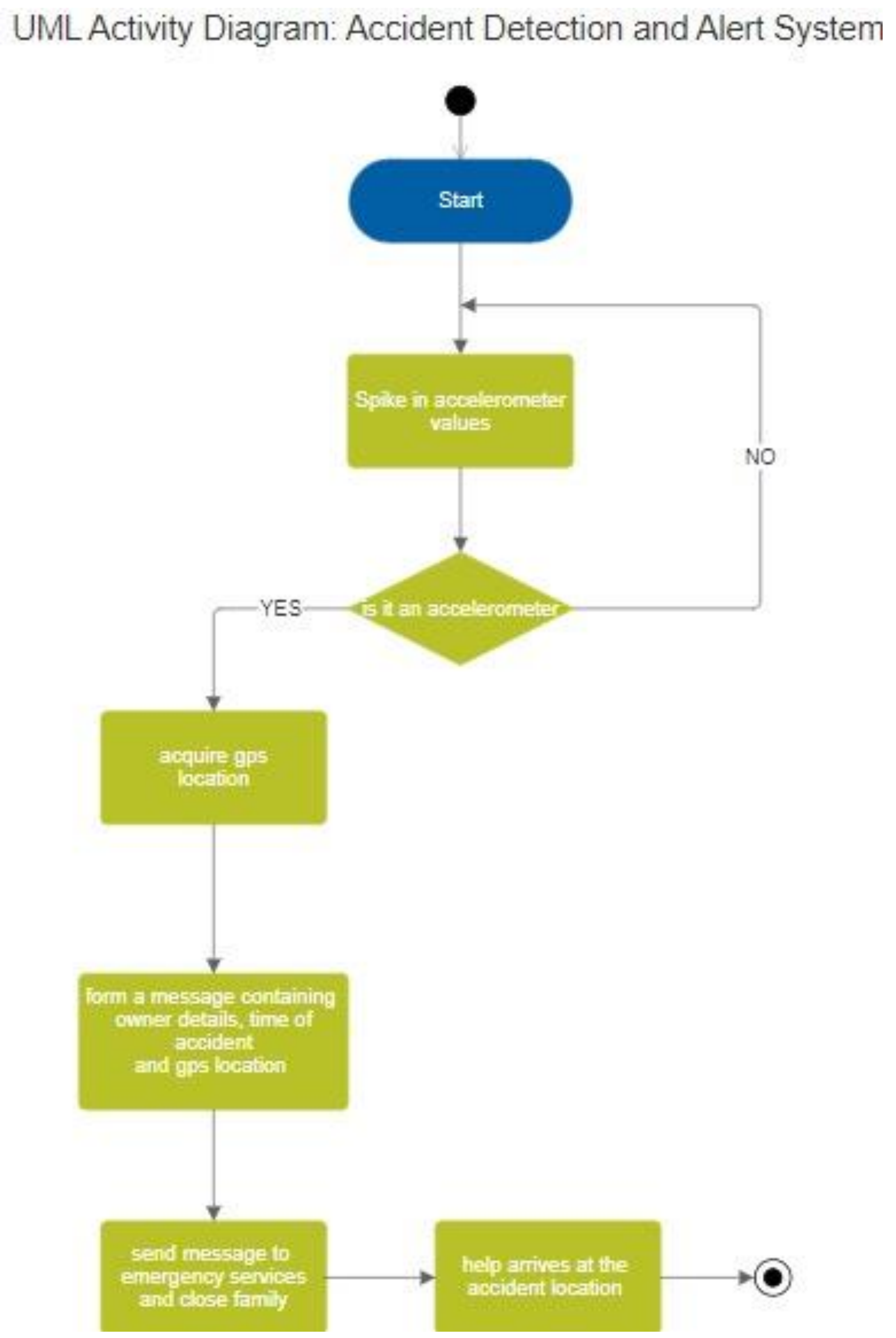


Generic Block Diagram

UML Activity Diagram: Accident Detection and Alert System



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Results & Discussion :

After going through the data collected, we can see that at certain data points the value of acceleration in one or more directions(X,Y,Z) shoots up dramatically .These values usually correlate to the point at where the vehicle is going through an abnormal condition like a crash, being hit by another vehicle etc. These values (found by analyzing the dataset) are used to set the boundary values to figure out whether the spike in the values was an accident or just a false indication.

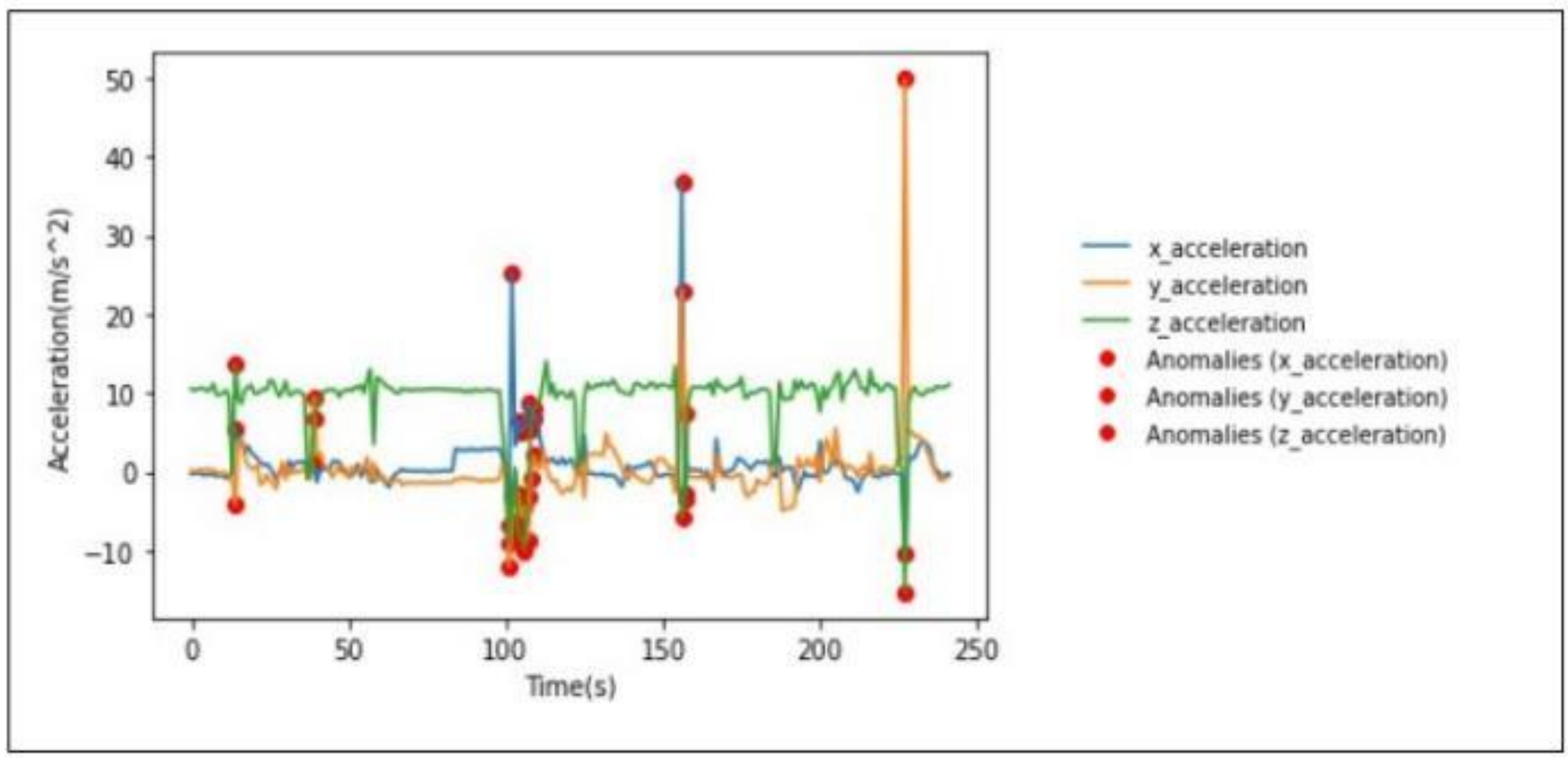

Figure 1.2: Acceleration experienced by Vehicle during Tumbling

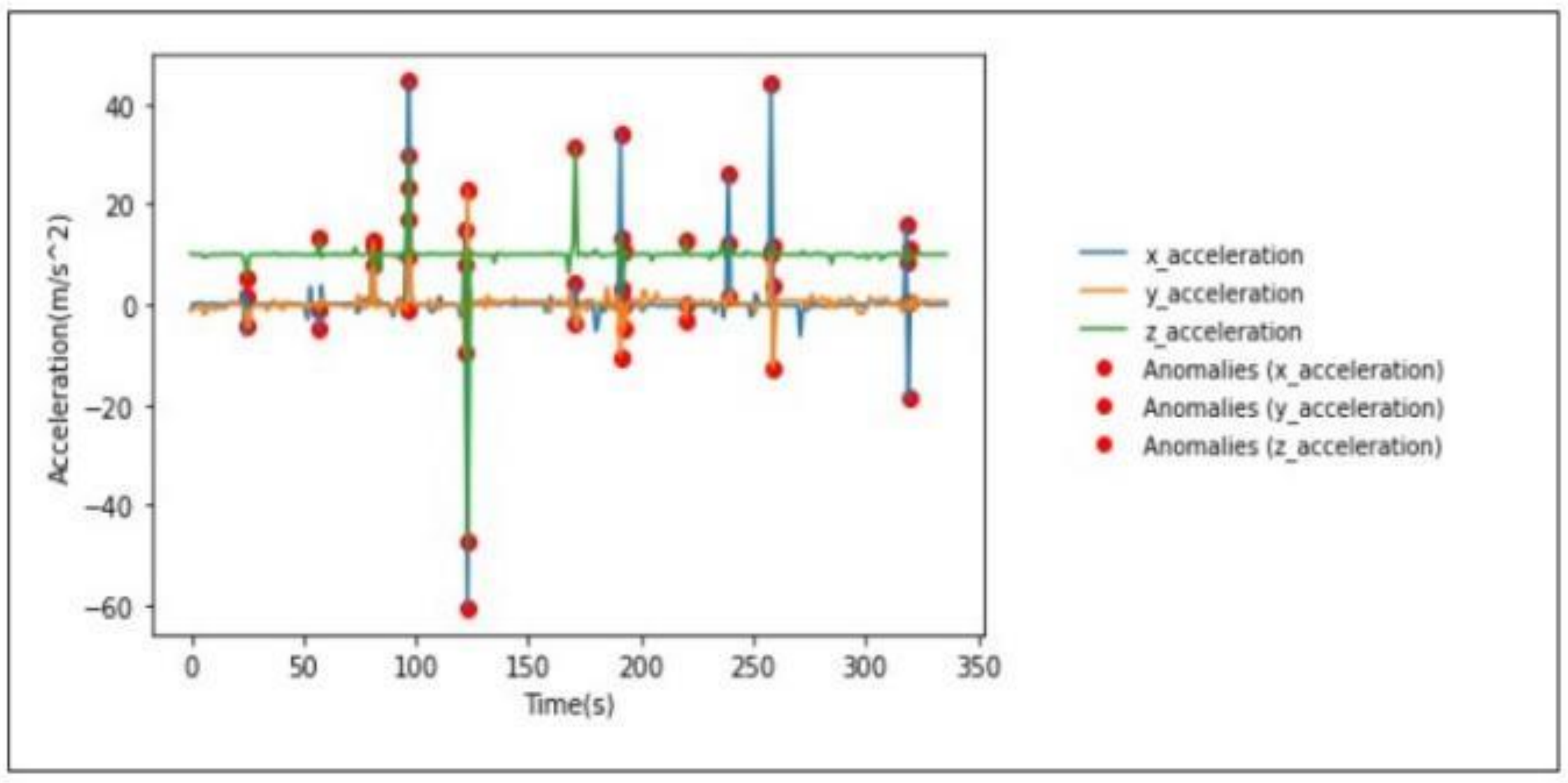

Figure 1.1: Acceleration experienced by Vehicle during Crash

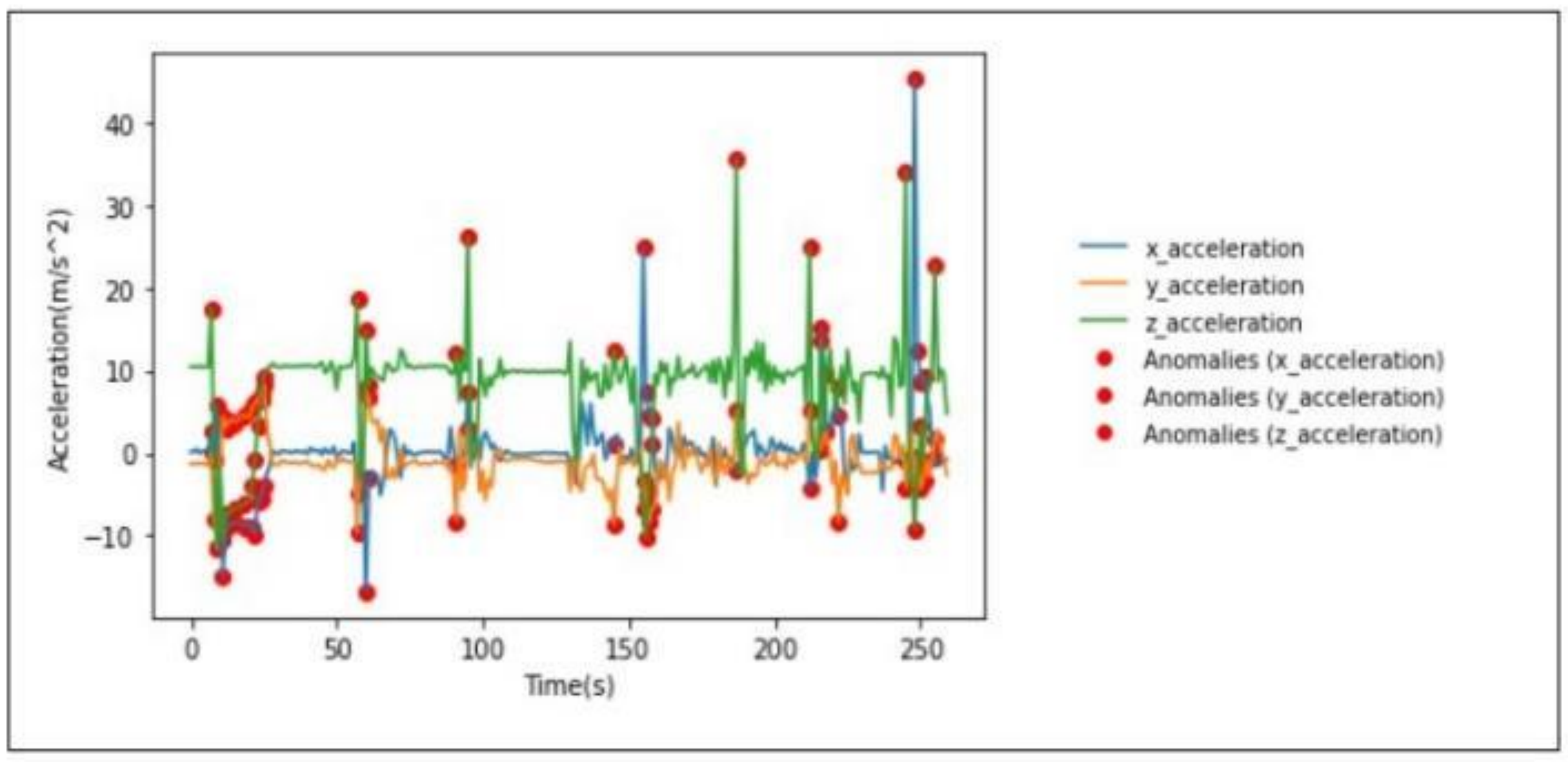

Figure 1.4: Acceleration experienced by Vehicle during Explosion

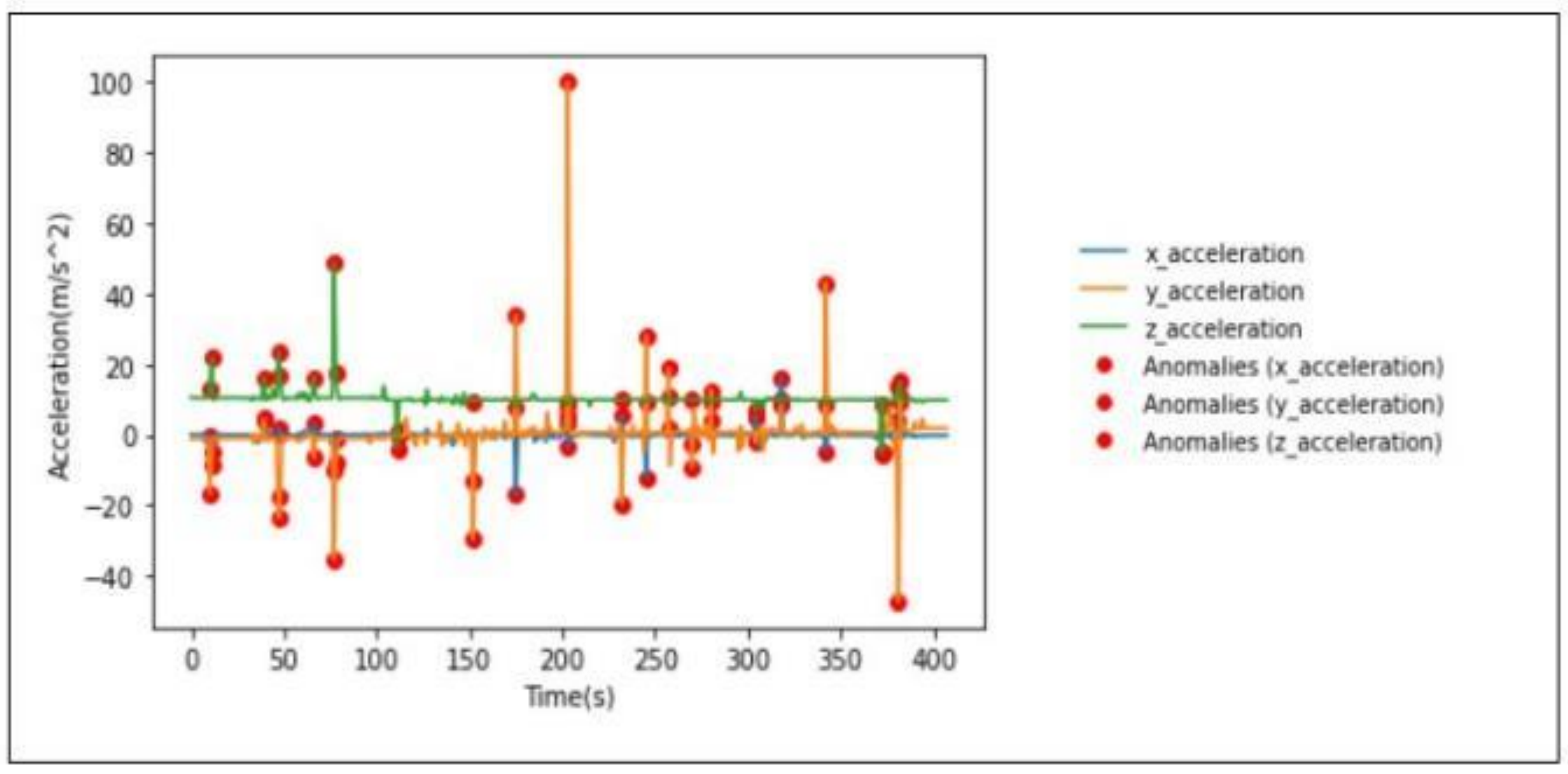

Figure 1.3: Acceleration experienced by Vehicle hit at sides

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Conclusions :

This project is a cutting-edge Accident Detection and Alert System in this initiative using technology to improve road safety. The system is built around an Arduino Uno as the processor, a SIM800 GSM module, for connectivity an ADXL345 3-axis accelerometer for motion sensing and a GPS module for location tracking. This setup ensures reliability and efficiency in operation. Ultimately this project aims to enhance the safety of driving environments by offering a solution that can be adapted to different vehicle types and settings. By reducing emergency response times and enhancing the precision of accident notifications our Accident Detection and Alert System represents a step, in automotive safety technology.

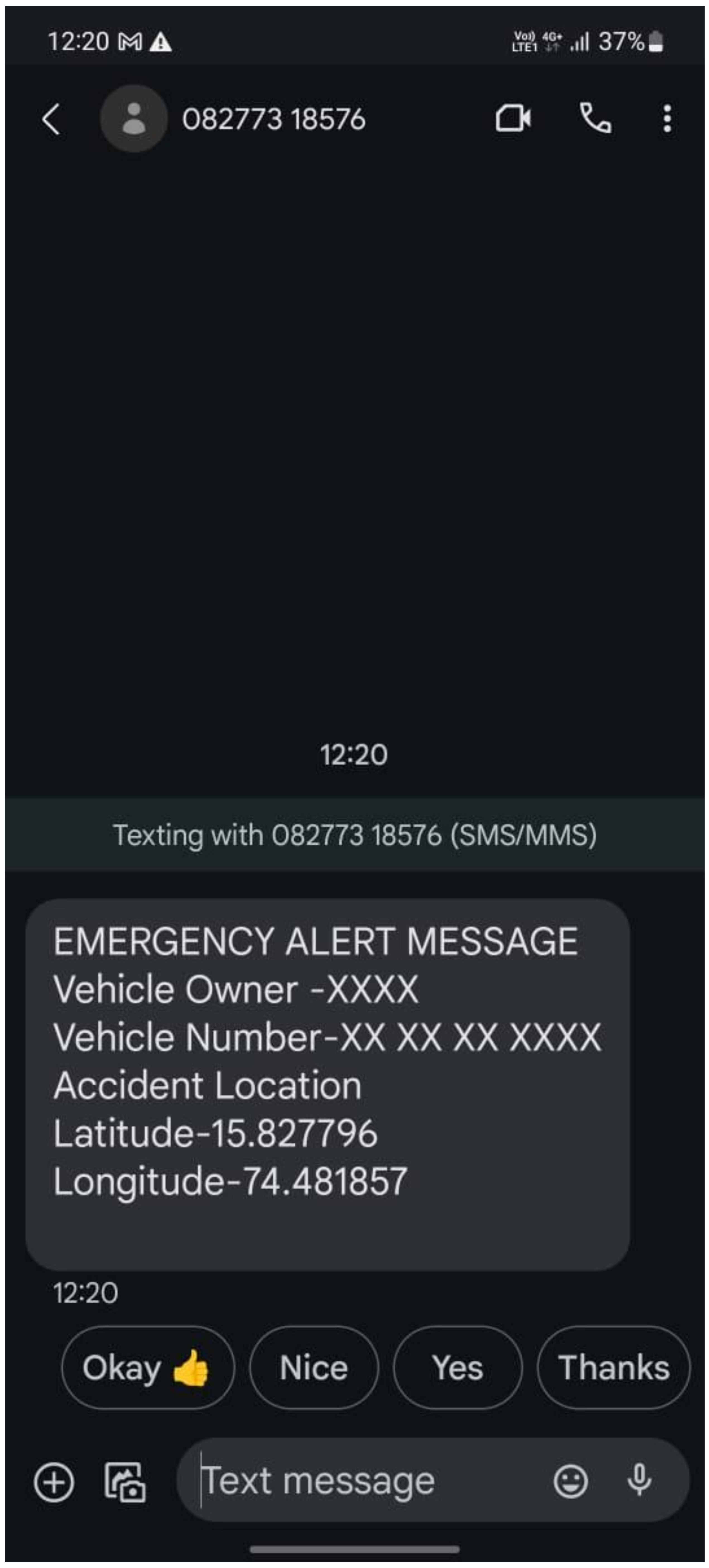

Figure 1.5: Emergency Alert Message

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